

lean4ml

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Theorem 1 (Smale 1958). *There is a homotopy of immersions of S^2 into $\mathbb{R}^3$ from the inclusion map to the antipodal map $a : q \mapsto -q$.*

Proof. This obviously follows from what we did so far. $\square$

TaylorAndSmoothness

Throughout this section, E is a real Hilbert space, $f : E \rightarrow \mathbb{R}$, $s \subseteq E$, and $L \in \mathbb{R}_{\geq 0}$ (Lean: `NNReal`).

Definition 2 (L -smoothness on a set). We say that f is L -smooth on s if

$$\|\nabla f(x) - \nabla f(y)\| \leq L\|x - y\|, \quad \forall x, y \in s.$$

In Lean: `LSmoothOn f L s := LipschitzOnWith L (fun x => gradient f x) s`.

Definition 3 (Global L -smoothness). A function is globally L -smooth if ∇f is L -Lipschitz on all of E . In Lean: `LSmoothfn f L := LipschitzWith L (fun x => gradient f x)`.

Lemma 4 (Global smoothness is smoothness on `Set.univ`). *In Lean:*

$$LSmoothfn f L \iff LSmoothOn f L \text{ Set.univ}.$$

This is a simp lemma using `lipschitzOnWith_univ`.

Lemma 5 (Monotonicity under set inclusion). *If $T \subseteq S$ and f is L -smooth on S , then f is L -smooth on T .*

Definition 6 (Quadratic upper bound model). The quadratic upper model on s is:

$$\forall x, y \in s, \quad f(y) \leq f(x) + \langle \nabla f(x), y - x \rangle + \frac{L}{2} \|y - x\|^2.$$

Definition 7 (Hessian). The Hessian is the Fréchet derivative of the gradient map:

$$\nabla^2 f(x) := D(\nabla f)(x).$$

In Lean: `hessian f x := fderiv (fun y => gradient f y) x`.

Lemma 8 (Derivative operator-norm bound from Lipschitz). *If g is L -Lipschitz on a set s containing a neighborhood of x and g is differentiable at x , then $\|Dg(x)\| \leq L$. (Lean: `HasFDerivAt.le_of_lipschitzOn`.)*

Lemma 9 (Quadratic form controlled by operator norm). *For $A : E \rightarrow \mathcal{L}[\mathbb{R}]E$ and $p \in E$:*

$$\langle Ap, p \rangle \leq \|A\| \|p\|^2.$$

Lean uses `real_inner_le_norm` and `ContinuousLinearMap.le_opNorm`.

Lemma 10 (HasGradient everywhere for C^1). *If `ContDiff 1 f`, then for every z we have `HasGradientAt f (gradient f z) z`.*

Lemma 11 (Norm Lipschitz bound for the gradient). *If f is L -smooth on s and $x, y \in s$, then $\|\nabla f(x) - \nabla f(y)\| \leq L\|x - y\|$. Lean rewrites the `edist` statement of `LipschitzOnWith`.*

Lemma 12 (Continuity of $t \mapsto (Df(x+tp))p$). *If f is C^1 , then $t \mapsto (Df(x+tp))p$ is continuous.*

Lemma 13 (Continuity of $t \mapsto \langle \nabla f(x+tp), p \rangle$). *Lean rewrites $(Df(x+tp))p$ into $\langle \nabla f(x+tp), p \rangle$ using `HasGradientAt.fderiv_apply`.*

Lemma 14 (Integrability of the line integrand). *The integrand is **IntervalIntegrable** on any interval by continuity.*

Lemma 15 (Segment norm identity). *For $\gamma \geq 0$, $\|(x + \gamma(y - x)) - x\| = \gamma\|y - x\|$.*

Lemma 16 (Line derivative). *The map $\tau \mapsto x + \tau \cdot p$ has derivative p at all τ .*

Lemma 17 (Chain rule along a line). *If f has Fréchet derivative at $x + tp$, then $\tau \mapsto f(x + \tau p)$ has derivative $(Df(x + tp))p$ at $\tau = t$.*

Lemma 18 (Integral split-off constant term in an inner product). *Lean lemma used to rewrite $\int \langle g(t) - c, w \rangle = \int \langle g(t), w \rangle - \int \langle c, w \rangle$.*

Lemma 19 (Monotonicity of interval integrals (continuous case)). *If $\varphi \leq \psi$ on $[a, b]$ with $a \leq b$ and both are continuous, then $\int_a^b \varphi \leq \int_a^b \psi$.*

Lemma 20 (Inner-product linearity rewrite). $\langle u, w \rangle - \langle v, w \rangle = \langle u - v, w \rangle$.

Lemma 21 (Convex segment stays in a convex set). *If s is convex and $x, y \in s$, then $x + t(y - x) \in s$ for $t \in [0, 1]$.*

Theorem 22 (First-order Taylor formula along a line (vector-valued)). *If $f : E \rightarrow F$ is C^1 , then*

$$f(x + p) - f(x) = \int_0^1 (Df(x + tp))p \, dt.$$

Lean uses `intervalIntegral.integral_eq_sub_of_hasDerivAt`.

Theorem 23 (First-order Taylor formula along a line). *If $f \in C^1$, then*

$$f(x + p) - f(x) = \int_0^1 \langle \nabla f(x + tp), p \rangle \, dt.$$

Theorem 24 (Second-order Taylor formula for the gradient). *If $f \in C^2$, then*

$$\nabla f(x + p) - \nabla f(x) = \int_0^1 \nabla^2 f(x + tp) p \, dt.$$

Lemma 25 (Pointwise Lipschitz-gradient bound). *Under L -smoothness, for $\gamma \geq 0$ and the segment point in s ,*

$$\langle \nabla f(x + \gamma(y - x)) - \nabla f(x), y - x \rangle \leq L\gamma\|y - x\|^2.$$

Lemma 26 (Non-parametric corollary ($\gamma = 1$)). *If $x, y \in s$ then $\langle \nabla f(y) - \nabla f(x), y - x \rangle \leq L\|y - x\|^2$.*

Theorem 27 (Quadratic upper bound from smoothness on a convex set). *If $f \in C^1$, f is L -smooth on convex s , and $x, y \in s$, then*

$$f(y) \leq f(x) + \langle \nabla f(x), y - x \rangle + \frac{L}{2}\|y - x\|^2.$$

Proof. This proof follows the Lean file `Optimization/LSmooth.lean` line-by-line. Let $p := y - x$.

1. Apply Theorem 23 to obtain $f(y) - f(x) = \int_0^1 \langle \nabla f(x + tp), p \rangle dt$.
2. Rewrite $f(y) - f(x) - \langle \nabla f(x), p \rangle$ as an integral of $\langle \nabla f(x + tp) - \nabla f(x), p \rangle$ using Lemma 18.
3. Use convexity via Lemma 21 to ensure $x + tp \in s$ for $t \in [0, 1]$.
4. Bound the integrand pointwise using Lemma 25 with $\gamma = t$, then integrate the bound with Lemma 19.
5. Evaluate $\int_0^1 Lt dt = L/2$ using `integral_id` and close by `linarith`.

□

HessianVsSmoothness

Theorem 28 (Hessian operator-norm bound $\Rightarrow$ smoothness on `univ`). *If ∇f is differentiable everywhere with derivative $\nabla^2 f(x)$, and $\|\nabla^2 f(x)\| \leq L$ for all x , then f is L -smooth on `Set.univ`.*

Proof. Lean constructs `hDiff : x, DifferentiableAt (fun y => gradient f y) x` from the `HasFDerivAt` hypothesis, rewrites the bound as an `nnorm` bound on `fderiv`, and applies the Mathlib lemma `lipschitzWith_of_nnorm_fderiv_le`. Finally it uses `lipschitzOnWith_univ.2`.

□

Theorem 29 (Smoothness on `univ` $\Rightarrow$ Hessian quadratic-form upper bound). *If f is L -smooth on `Set.univ`, then $\langle \nabla^2 f(x)p, p \rangle \leq L\|p\|^2$ for all x, p .*

Proof. Lean first obtains $\|\nabla^2 f(x)\| \leq L$ from the Lipschitz hypothesis using Lemma 8, then applies Lemma 9 and finishes via `gcongr`.

□

NecessaryConditions

Theorem 30 (Local minimum $\Rightarrow$ gradient vanishes). *If $f \in C^1$ and x is a local minimizer (Lean: `IsLocalMin f x`), then $\nabla f(x) = 0$.*

Proof. Lean:

1. obtain `HasGradientAt f (gradient f x) x` from Lemma 10;
2. apply `IsLocalMin.hasFDerivAt_eq_zero` to get a dual-space derivative equality;
3. evaluate this equality at $\nabla f(x)$ to deduce $\langle \nabla f(x), \nabla f(x) \rangle = 0$;
4. conclude $\nabla f(x) = 0$ by `inner_self_eq_zero.mp`.

□

Lemma 31 (Hessian continuity along a line). *If $f \in C^2$, then $t \mapsto \nabla^2 f(x + tp)p$ is continuous.*

Lemma 32 (Scaling of the Hessian quadratic form). $\langle \nabla^2 f(x)(\alpha p), \alpha p \rangle = \alpha^2 \langle \nabla^2 f(x)p, p \rangle$.

Lemma 33 (Gradient as integral of Hessian along a line). *If $f \in C^2$ and $\nabla f(x) = 0$, then $\nabla f(x + tp) = \int_0^t \nabla^2 f(x + sp)p ds$.*

Lemma 34 (Exact second-order Taylor expansion (double integral)). *If $f \in C^2$ and $\nabla f(x) = 0$, then*

$$f(x + p) - f(x) = \int_0^1 \int_0^t \langle \nabla^2 f(x + sp)p, p \rangle ds dt.$$

Theorem 35 (Local minimum $\Rightarrow$ Hessian PSD). *If $f \in C^2$ and x is a local minimizer, then $\langle \nabla^2 f(x)p, p \rangle \geq 0$ for all p .*

Proof. Lean proof strategy:

1. reduce to the case $\nabla f(x) = 0$ via Theorem 30;
2. assume $\langle \nabla^2 f(x)p, p \rangle < 0$ and set $H(u) := \langle \nabla^2 f(x + up)p, p \rangle$;
3. use continuity (Lemma 31) to find $\delta > 0$ so that $H(u) < H(0)/2$ on $\text{ball}(0, \delta)$;
4. use local minimality to show $f(x + \alpha p) - f(x) \geq 0$ eventually as $\alpha \downarrow 0$;
5. use the exact Taylor double-integral formula (Lemma 34) and monotonicity of integrals to show $f(x + \alpha p) - f(x) < 0$ eventually for $\alpha > 0$ small, contradiction.

□

Sufficient Conditions

Definition 36 (Strict local minimizer). Lean definition: $\exists U \in \mathcal{N}(x), \forall y \in U, y \neq x \rightarrow f(x) < f(y)$.

Lemma 37 (Taylor lower bound from uniform Hessian lower bound). *Under a uniform bound $\varepsilon \|p\|^2 \leq \langle \nabla^2 f(y)p, p \rangle$ on a ball around x and assuming $\nabla f(x) = 0$, Lean proves $(\varepsilon/2)\|p\|^2 \leq f(x+p) - f(x)$ for $\|p\| < \rho$ by integrating the pointwise bound inside the double-integral remainder formula.*

Proof. Lean proof:

1. start from the identity in Lemma 34;
2. prove that $x + sp$ stays in the ball for $s \in [0, 1]$ using norm estimates (`norm_smul`, `gcongr`);
3. apply `intervalIntegral_mono_of_continuous` twice (inner integral in s , outer in t);
4. evaluate $\int_0^1 \varepsilon \|p\|^2 t dt = (\varepsilon/2)\|p\|^2$ using `integral_id` and `ring`.

□

Lemma 38 (Coercivity from positive definiteness (finite-dimensional)). *In finite dimensions, a positive definite continuous linear map H satisfies $\exists \varepsilon > 0, \forall p, \varepsilon \|p\|^2 \leq \langle Hp, p \rangle$. Lean uses compactness of the sphere and `IsCompact.exists_isLeast`.*

Lemma 39 (Continuity of Hessian map). *If $f \in C^2$, then $y \mapsto \nabla^2 f(y)$ is continuous (Lean: `continuous_fderiv` applied to ∇f).*

Lemma 40 (Positive definite Hessian $\Rightarrow$ uniform lower bound on a ball). *In finite dimensions, if $\nabla^2 f(x)$ is positive definite, then there exist $\varepsilon > 0$ and $\rho > 0$ such that $\varepsilon \|p\|^2 \leq \langle \nabla^2 f(y)p, p \rangle$ for all $y \in B(x, \rho)$ and all p .*

Proof. Lean proof:

1. use Lemma 38 at $H = \nabla^2 f(x)$ to get ε_0 ;
2. use continuity of $y \mapsto \nabla^2 f(y)$ to pick ρ so that $\|\nabla^2 f(y) - \nabla^2 f(x)\| < \varepsilon_0/2$;
3. bound $|\langle (\nabla^2 f(y) - \nabla^2 f(x))p, p \rangle|$ by $\|\nabla^2 f(y) - \nabla^2 f(x)\| \|p\|^2$ (operator norm estimate), then conclude the desired lower bound by rearranging with `linarith`.

□

Theorem 41 (Strict local minimizer from uniform Hessian lower bound). *If $f \in C^2$, $\nabla f(x) = 0$, and the Hessian has a uniform positive lower bound on a ball around x , then x is a strict local minimizer.*

Proof. Lean sets $U = B(x, \rho)$. For $y \in U$ with $y \neq x$, set $p = y - x$ so that $y = x + p$ and $\|p\| < \rho$. Lemma 37 yields $(\varepsilon/2)\|p\|^2 \leq f(y) - f(x)$. Since $p \neq 0$, the right-hand side is strictly positive, hence $f(x) < f(y)$. □

Theorem 42 (Second-order sufficient condition (finite-dimensional)). *If $f \in C^2$, $\nabla f(x) = 0$, and $\nabla^2 f(x)$ is positive definite, then x is a strict local minimizer.*

Proof. Lean applies Lemma 40 to obtain ε, ρ and the uniform Hessian lower bound, then applies Theorem 41. □

GradientFlow

Throughout this section, E is a real Hilbert space, $f : E \rightarrow \mathbb{R}$, and ∇f denotes the (Fréchet) gradient in the Hilbert space structure.

Definition 43 (Gradient flow on a bounded interval). A *bounded-time* gradient flow is a certificate that the ODE

$$\dot{x}(t) = -\nabla f(x(t)), \quad x(0) = x_0$$

has a solution on $[0, T]$. In Lean, a `GradientFlow f x T` is a structure with fields:

- `trajectory : ℝ → E`,
- `init : trajectory 0 = x`,
- `hasDerivAt : t → Icc 0 T, HasDerivAt trajectory (-(gradient f (trajectory t))) t`.

Lemma 44 (Initial value). *For $\varphi : \text{GradientFlow } f \ x \ T$, we have $\varphi(0) = x_0$.*

Lemma 45 (Differentiability of the trajectory on $[0, T]$). *For $t \in [0, T]$, the map $t \mapsto \varphi(t)$ is differentiable at t . (Lean: `DifferentiableAt .trajectory t`.)*

Lemma 46 (Derivative equals negative gradient). *For $t \in [0, T]$,*

$$\varphi'(t) = -\nabla f(\varphi(t)).$$

(Lean: `deriv .trajectory t = -(gradient f (.trajectory t))`.)

Lemma 47 (Energy derivative along the flow). *If $f \in C^1$ (Lean: `ContDiff 1 f`) and φ is a gradient flow, then for $t \in [0, T]$*

$$\frac{d}{dt} f(\varphi(t)) = -\|\nabla f(\varphi(t))\|^2.$$

Lean proof outline:

- apply the chain rule `HasFDerivAt.comp_hasDerivAt` to $f \circ \varphi$;
- rewrite `fderiv` in terms of `gradient` by `simp [gradient]`;
- substitute the ODE via Lemma 46;
- finish with `inner_neg_right` and `real_inner_self_eq_norm_sq`.

Lemma 48 (Energy is non-increasing). *If $0 \leq r \leq s \leq T$, then*

$$f(\varphi(s)) \leq f(\varphi(r)).$$

Lean proof:

- argue by contradiction;
- apply the mean value theorem in the form `exists_deriv_eq_slope`;
- rewrite the derivative using Lemma 47;
- conclude with `nlinarith` and the nonnegativity of norms.

Definition 49 (Piecewise trajectory). The piecewise trajectory that follows φ_1 up to time T and then φ_2 (shifted) is

$$\Phi(t) := \begin{cases} \varphi_1(t), & t \leq T, \\ \varphi_2(t - T), & t > T. \end{cases}$$

(Lean: `if t ≤ T then φ₁ t else φ₂ (t - T)`.)

Definition 50 (Extend (chain) two gradient flows). Given a flow φ on $[0, T]$ and a flow ψ on $[0, \delta]$ starting at $\varphi(T)$, Lean constructs a flow on $[0, T + \delta]$ by the piecewise trajectory. The derivative condition is proved by trichotomy on $t < T$, $t = T$, and $t > T$ and by using `HasDerivWithinAt.union` at the junction time.

Definition 51 (Restrict a flow to a smaller time interval). A flow on $[0, S]$ restricts to one on $[0, T]$ for $T \leq S$ by reusing the same trajectory and weakening the domain proof.

Lemma 52 (Gradient map is continuous). *If $f \in C^1$, then ∇f is continuous. (Lean: proof is `unfold gradient; fun_prop`.)*

Lemma 53 (Norm bound from a Lipschitz gradient). *If ∇f is L -Lipschitz, then*

$$\|\nabla f(x)\| \leq \|\nabla f(y)\| + L\|x - y\|.$$

Lean uses `LipschitzWith.dist_le_mul` (after rewriting `dist` as $\|\cdot\|$) and then `linarith` with `norm_sub_norm_le`.

Theorem 54 (Local gradient flow existence). *If $f \in C^1$ and ∇f is globally L -Lipschitz, then for any x_0 there exists $\delta > 0$ and a gradient flow on $[0, \delta]$ starting at x_0 . Lean proof applies Picard–Lindelöf via `IsPicardLindelof.exists_eq_forall_mem_Icc_hasDerivWithinAt` to the vector field $x \mapsto -\nabla f(x)$, with explicit constants.*

Definition 55 (Concrete local flow constructor). Lean also builds an explicit flow on the explicit time step

$$\delta := \frac{1}{4L + 4},$$

by repeating the Picard–Lindelöf construction with these fixed constants.

Theorem 56 (Uniform local existence). *There exists $\delta > 0$ (in fact $\delta = 1/(4L + 4)$) such that for every x_0 there is a gradient flow on $[0, \delta]$ starting from x_0 .*

Lemma 57 (Iterated extension). *From a hypothesis x , `Nonempty (GradientFlow f x)` and $\delta > 0$, Lean builds a flow on $[0, n\delta]$ by induction on n .*

Theorem 58 (Gradient flow existence on $[0, T]$). *If $f \in C^1$, ∇f is globally L -Lipschitz, and $T > 0$, then for any x_0 there exists a gradient flow on $[0, T]$ starting at x_0 . Lean takes $n := \lceil T/\delta \rceil_+ + 1$ so that $T \leq n\delta$, then restricts.*

Proof. Lean proof steps:

1. obtain $\delta > 0$ and a uniform local existence function from Theorem 56;
2. apply Lemma 57 with $n := \lceil T/\delta \rceil_+ + 1$ to obtain a flow on $[0, n\delta]$;
3. restrict to $[0, T]$ using Definition 51 and the ceiling inequality (Lean uses `Nat.le_ceil` and `linarith`).

□

Definition 59 (Global gradient flow). A `GradientFlowGlobal f x` is a trajectory defined for all $t \in \mathbb{R}$ with the ODE holding for all $t \geq 0$: `hasDerivAt : t, 0 ≤ t → HasDerivAt trajectory (-(gradient f (trajectory t))) t`.

Definition 60 (Restriction of a global flow). A global flow restricts to a bounded flow on $[0, T]$ by reusing the same trajectory.

Lemma 61 (Uniqueness of bounded flows). *If ∇f is L -Lipschitz and $T \geq 0$, then two flows with the same initial condition coincide on $[0, T]$ (Lean: `Set.EqOn ... (Icc 0 T)`). Lean reduces to `ODE_solution_unique`.*

Lemma 62 (Consistency at overlaps). *For a family of flows $\varphi(n)$ on $[0, n + 1]$, any two indices agree at any time in both domains. Lean proves this by restricting the larger interval and applying uniqueness.*

Lemma 63 (Local agreement of the glued global trajectory). *Define $x(t) := (\varphi(\lceil t \rceil_+)).trajectory(t)$. For each fixed $t \geq 0$, Lean shows x is eventually equal near t to $(\varphi(\lceil t \rceil_+)).trajectory$, enabling transfer of derivative statements by `HasDerivAt.congr_of_eventuallyEq`.*

Theorem 64 (Global gradient flow existence for all $t \geq 0$). *If $f \in C^1$ and ∇f is globally L -Lipschitz, then for any x_0 there exists a global gradient flow defined for all $t \geq 0$.*

Proof. Lean proof:

1. for each $n \in \mathbb{N}$, pick a bounded flow $\varphi(n)$ on $[0, n + 1]$ using Theorem 58;
2. define the global trajectory by $x(t) := (\varphi(\lceil t \rceil_+)).trajectory(t)$;
3. for each fixed $t \geq 0$, use Lemma 63 to show local agreement of x with $(\varphi(\lceil t \rceil_+)).trajectory$ near t ;
4. transfer `HasDerivAt` at t from the bounded flow to x using `HasDerivAt.congr_of_eventuallyEq`.

□